

Seat No: _____

Enrollment No: _____

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech./ Int. Btech Winter 2025 - 26 Examination

Semester: 7 / 11

Date: 19/12/2025

Subject Code: 303105377

Time: 02:00pm to 04:30pm

Subject Name: Software Testing & Quality Assurance

Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1 is compulsory**, From Section B, **Q.1 is compulsory**.
3. Figures to the right indicate full marks.
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION-A

Q.1 Answer the following questions.	Marks	CO	BT
A. i) Define the term Software Quality and explain its importance in software development?	[06]	3	2
ii) Differentiate between Verification and Validation with suitable examples?		3	2
B. i) Explain the concept of Software Configuration Items (SCIs) with an example?	[06]	3	2
ii) Describe the primary goal of Risk Identification in a software project?		3	2
Q.2 A. Design a set of test cases for a user registration form using the Equivalence Partitioning and Boundary Value Analysis techniques?	[04]	4	3
B. Discuss the primary activities of Software Quality Assurance (SQA)?	[05]	3	2
OR			
B. Explain the V-Model and its relationship with testing activities at each phase?	[05]	4	2
Q.3 A. Analyze the role of Six Sigma in achieving process improvement within a software development organization?	[04]	3	4
B. Propose a Software Configuration Management (SCM) strategy for a new e-commerce application?	[05]	5	6
OR			
B. Explain the importance of ISO 9001:2000 as a quality standard for software projects?	[05]	3	2

SECTION-B

Q.1 Answer the following questions.			
A. i) List and explain any four objectives of Software Testing?	[06]	4	1
ii) Define Total Quality Management (TQM) and state its importance in software development?		3	2
B. i) Explain the purpose of Smoke Testing and Regression Testing with suitable examples?	[06]	4	2
ii) Differentiate between Unit, Integration, System, and Acceptance Testing (any two key differences?)		4	2
Q.2 A. Explain the concept of Cyclomatic Complexity and its use in White Box testing?	[04]	4	2

B. Discuss the various types of software risks and provide a strategy for mitigating each type? **[05] 5 3**

OR

B. Explain the concept of a RMMM Plan. Discuss its importance with an example from a software project? **[05] 5 1**

Q.3 A. How would you design a test plan for a web-based application? **[04] 4 6**

B. Analyze the specialized testing requirements for real-time systems? **[05] 4 4**

OR

B. Discuss the role of software metrics in managing and improving the testing process? **[05] 3 3**